

Track Generation Report

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Generated: 2026-05-30 14:03:17

Gate file: ref_gates_autoX.csv

Gate count: 52

Optimization parameters

method: L-BFGS-B

maxiter: 3

ftol: 1e-07

Point counts

objective function points: 364

path generation points: 156

Results

total optimization time: 31.159 s

total path length: 827.3463 m

Per-step time estimate (one measured objective evaluation)

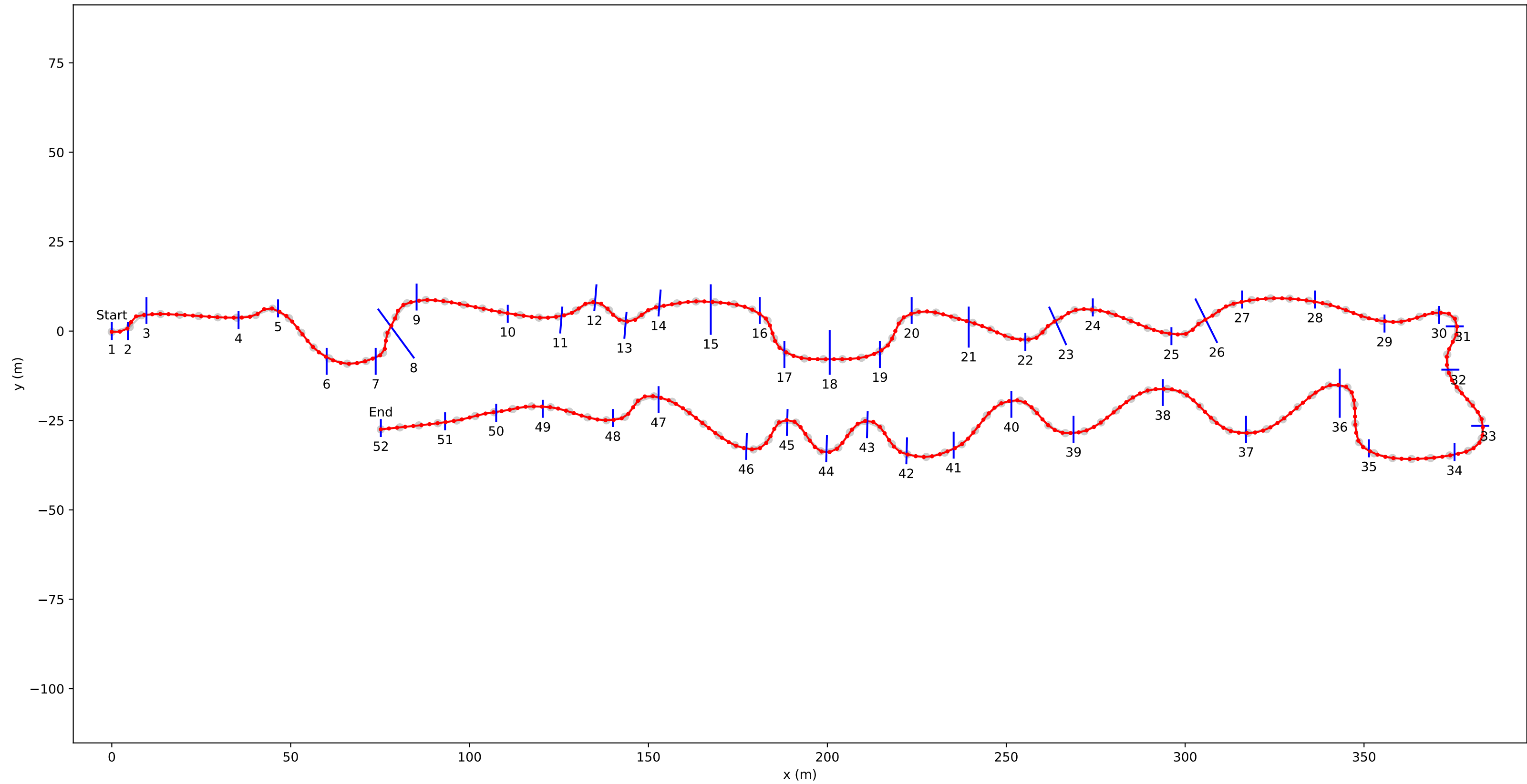
make spline from dofs: 8.77 ms

path length function: 0.60 ms

equidistant times generated: 0.59 ms

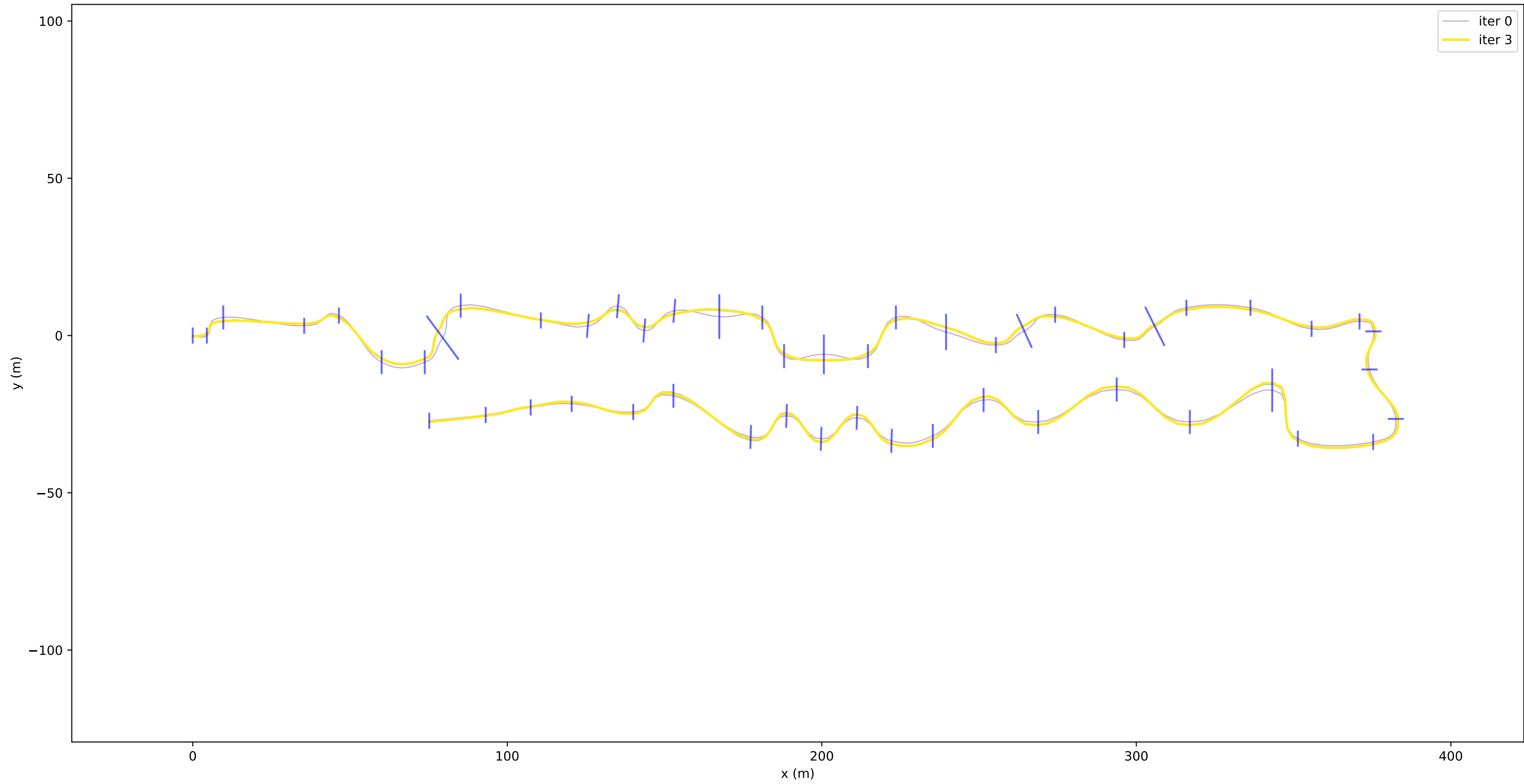
curvature objective function: 18.93 ms

Map plot: optimized path through gates



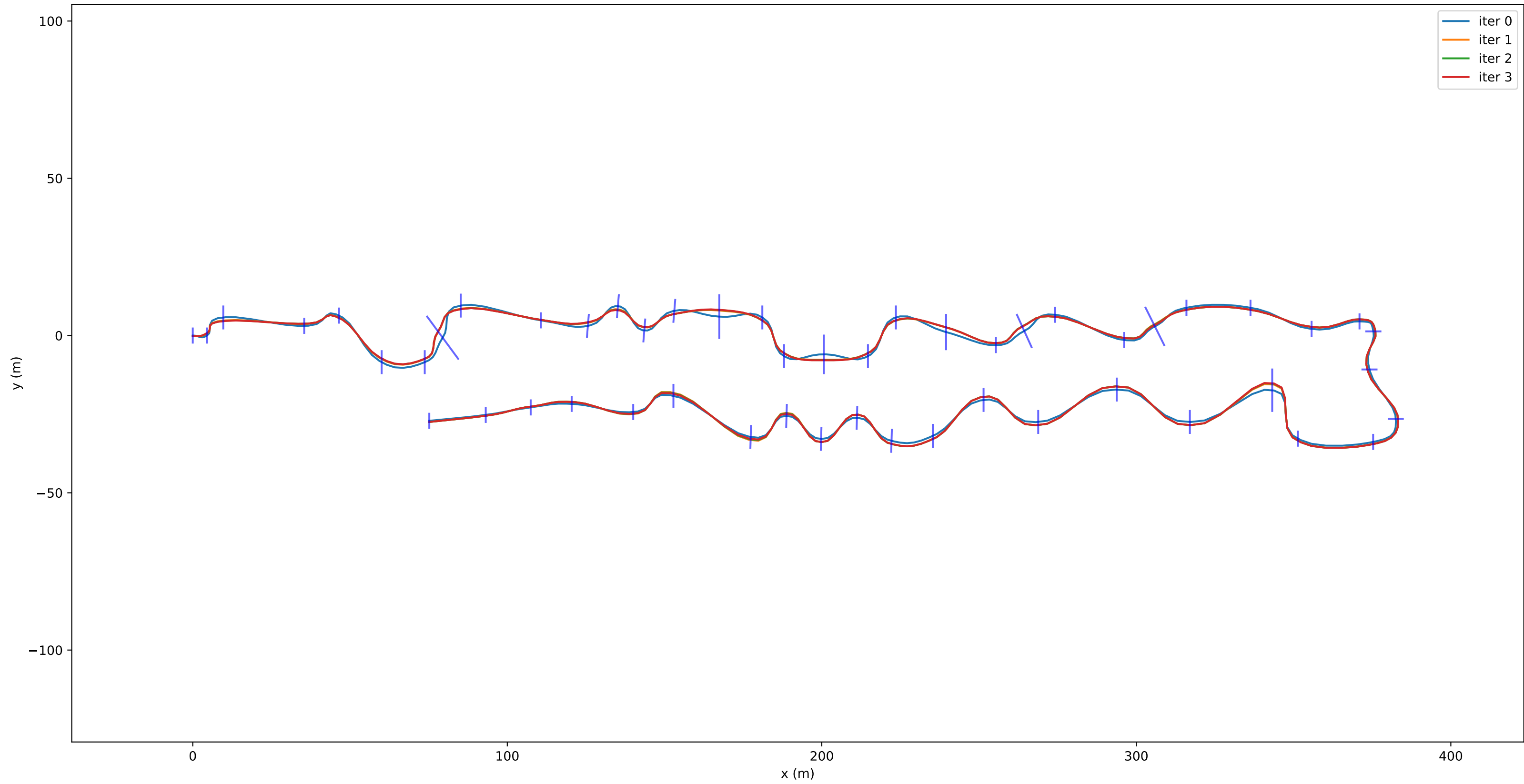
Optimization progression

Optimization progression (4 iterations)

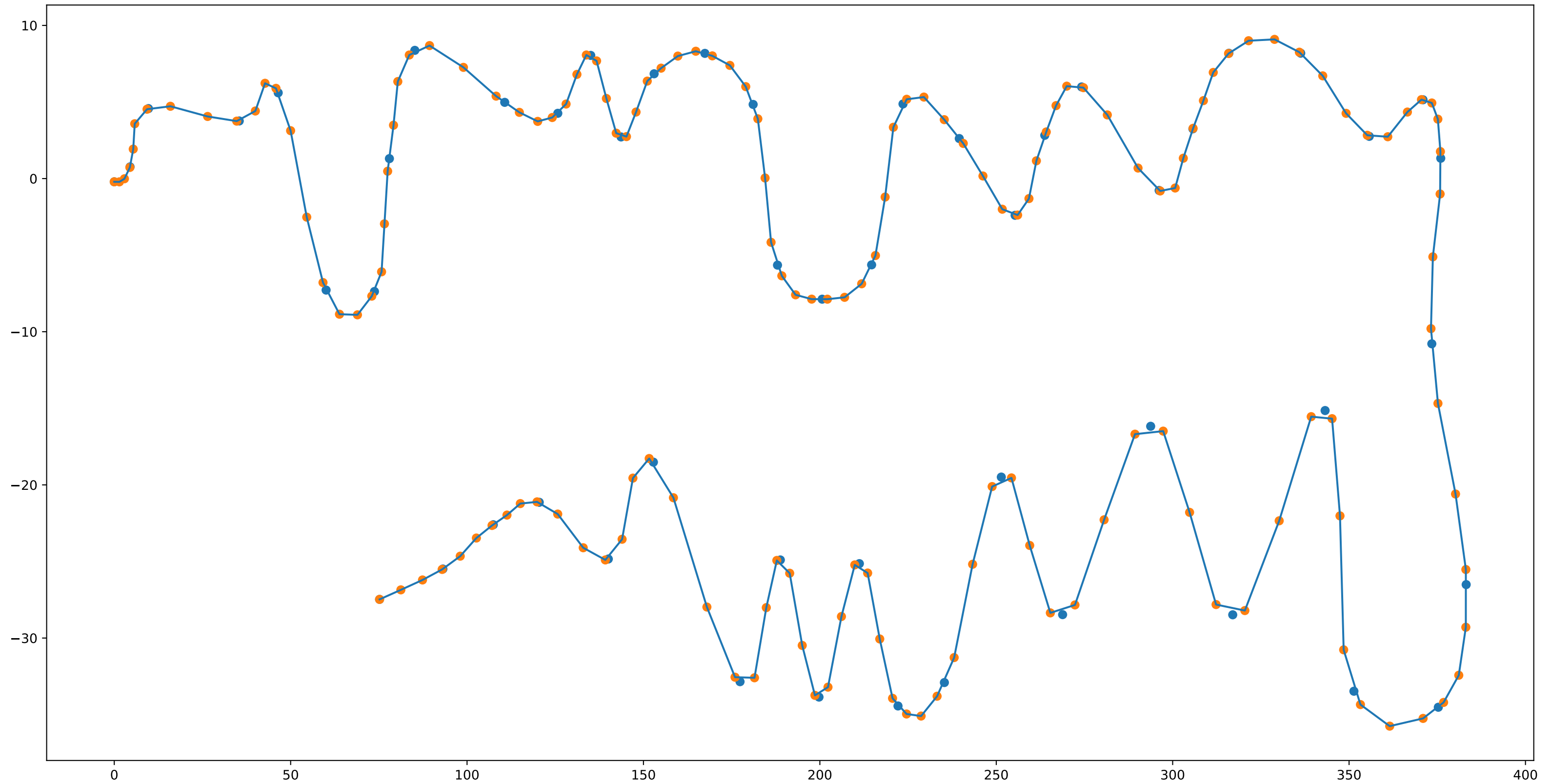


Equally spaced iterations

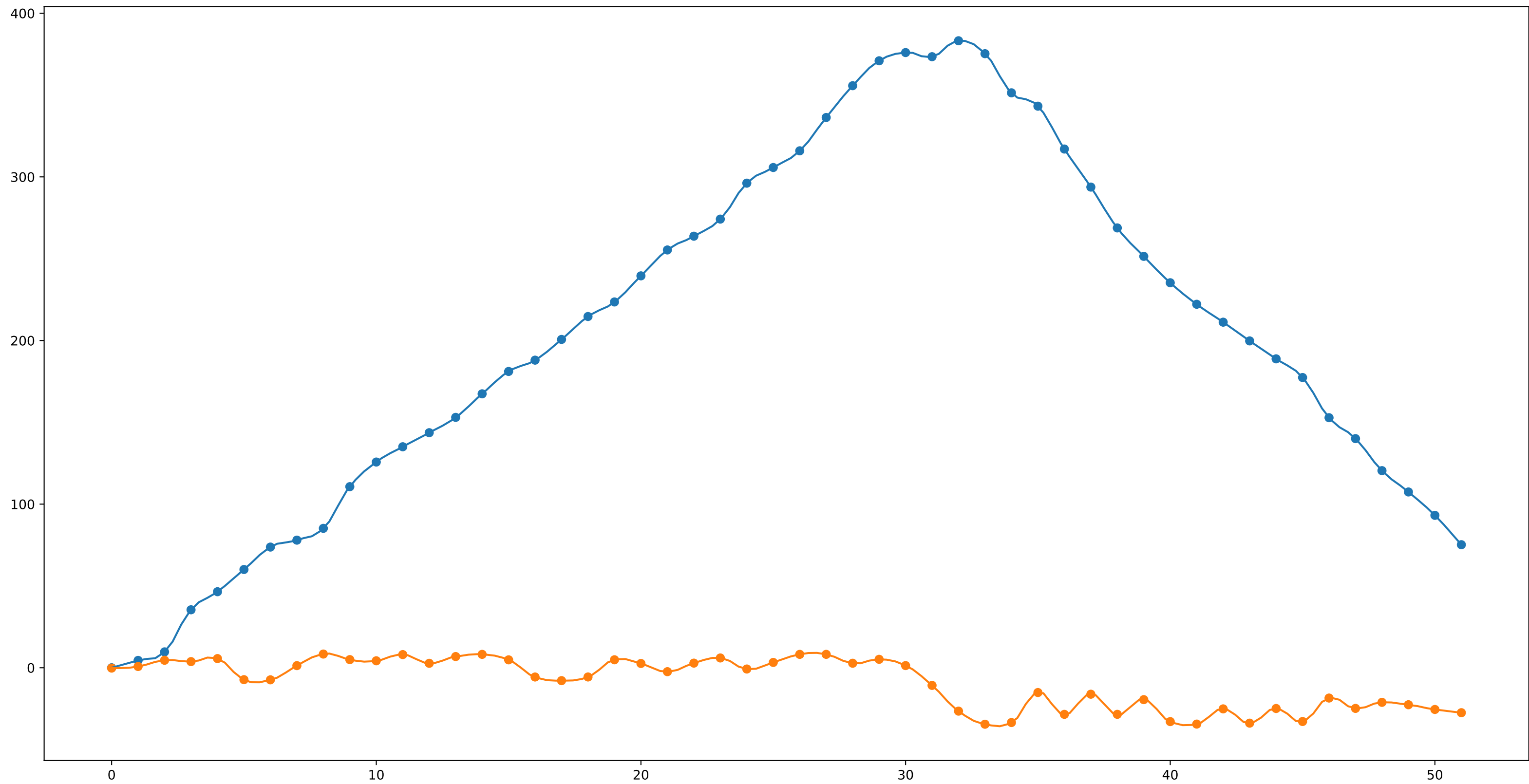
4 equally spaced iterations



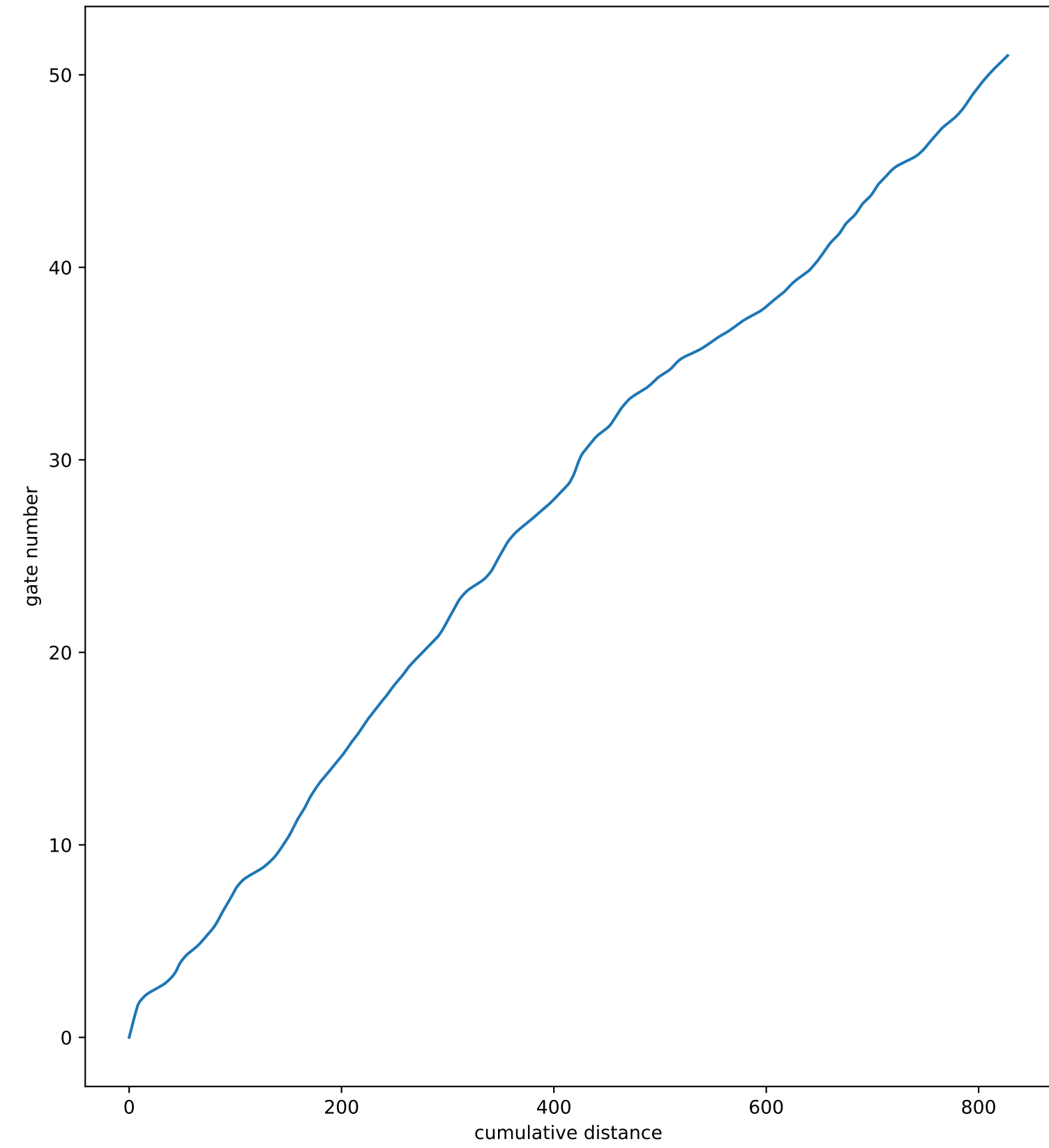
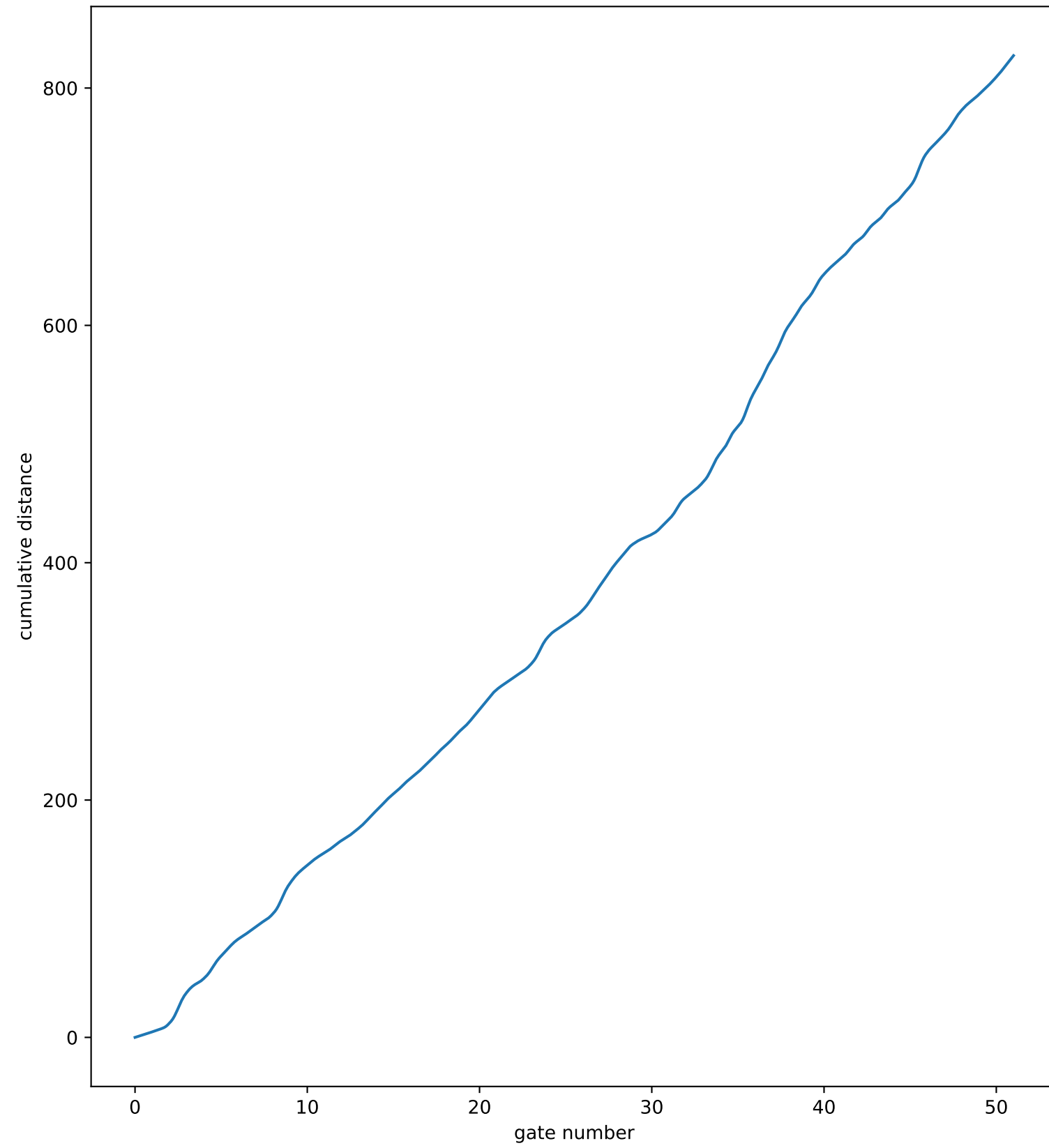
Spline debug: path and gate cross points



Spline debug: x(t) and y(t)



Path debug: cumulative distance vs gate number



Equidistant fit: distance to t mapping

